Arithmetic sequences

1 How is the common difference of an arithmetic sequence obtained?

2 Determine if the following sequences are arithmetic progressions:

a $3, 6, 12, 24, \ldots$

b $5, 7, 5, 7, \ldots$

3 Write down the next two terms of the following arithmetic sequences:

a $4, 8, 12, 16, \ldots$

b $6, 2, -2, -6, \ldots$

c $2, 3.5, 5, 6.5, \ldots$

d $-8, -\frac{23}{3}, -\frac{22}{3}, -7, \ldots$

4 Find the common difference of the following arithmetic sequences:

a $330, 280, 230, 180, \ldots$

b $-6, -\frac{39}{7}, -\frac{36}{7}, -\frac{33}{7}, \ldots$

5 For each of the following definitions of the nth term of a sequence:

i List the first four terms of the sequence.

ii Find the common difference.

a $T_n = 3n + 8$

b $T_n = 11 + (n - 1) \times 10$

c $T_n = -7 - 3(n - 1)$

6 Consider the arithmetic sequence $1.4, 2.3, 3.2, \ldots, 10.4$.

a Determine d, the common difference.

b Find n, the number of terms in the sequence.

7 Insert five terms in the arithmetic sequence which has -12 as its first term and 24 as its last term.

$$-12, \square, \square, \square, \square, \square, 24$$

8 For each of the following sequences, find the value of n:

a $0.9, 1.5, 2.1, \ldots$ where $T_n = 22.5$

b $2, 7, 12, \ldots$ where $T_n = 132$

c $2, -3, -8, \ldots$ where $T_n = -578$

d $5, \frac{17}{4}, \frac{7}{2}, \ldots$ where $T_n = -37$

9 For each of the following sequences:

i Determine d, the common difference.

ii State the expression for finding T_n, the nth term in the sequence.

iii Determine T_{10}, the 10th term in the sequence.

a $5, 12, 19, \ldots$

b $17, 16.2, 15.4, \ldots$

c $10, 3, -4, \ldots$

d $5, \frac{23}{4}, \frac{13}{2}, \ldots$

Recursive rules

10 Each term of a sequence is obtained by the previous term minus 6. The first term is 3. Write a recursive rule for T_n in terms of T_{n-1} and an initial condition for T_1 that defines this sequence.

11 Each term is obtained by increasing the previous term by 25. The first term is 30. Write a recursive rule for T_n in terms of T_{n-1} and an initial condition for T_1 for this sequence.

12 The first term of an arithmetic sequence is 2. The fifth term is 26.

a Find d, the common difference.

b Write a recursive rule for T_n in terms of T_{n-1} which defines this sequence and an initial condition for T_1.

13 Consider the arithmetic sequence with terms $T_3 = 14$ and $T_{12} = 59$.

a Find d, the common difference.

b Find the term T_1.

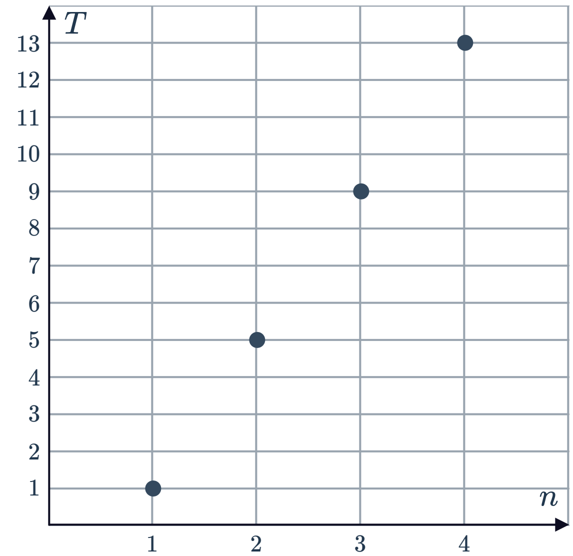
c Write a recursive rule for T_n in terms of T_{n-1} which defines this sequence and an initial condition for T_1.

Tables and graphs

14 The plotted points represent terms in an arithmetic sequence:

a Complete the table of values for the given points:

n	1	2	3	4
T_n				



b State the value of d, the common difference.

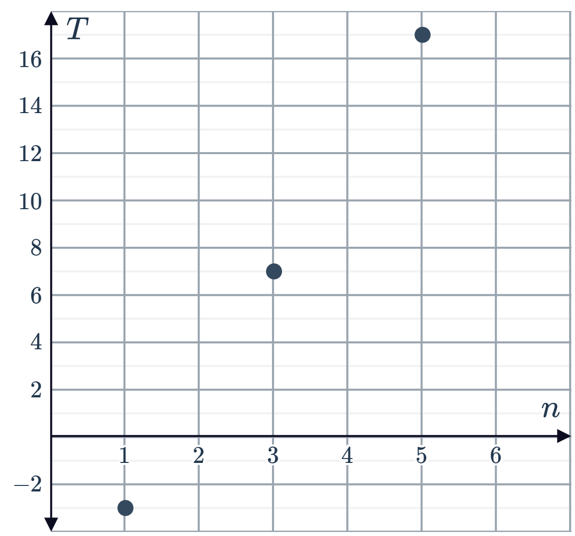
c Write a simplified expression for the general term, T_n.

d Find the 14th term of the sequence.

15 The plotted points represent terms in an arithmetic sequence:

a Complete the table of values for the given points:

n	1	3	5
T_n			



b State the value of d, the common difference.

c Write a simplified expression for the general term, T_n.

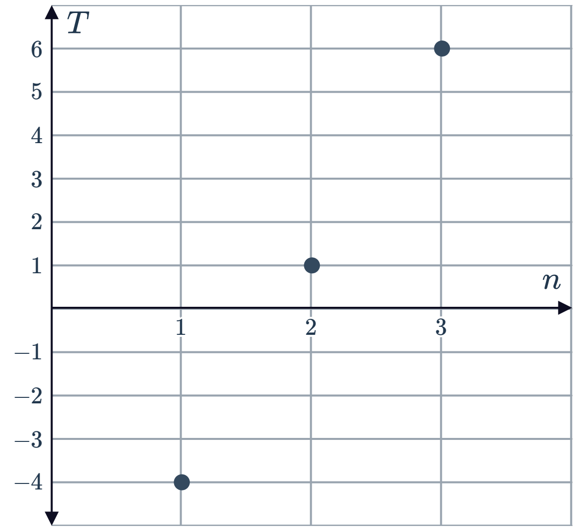
d Find the 18th term of the sequence.

16 The plotted points represent terms in an ari[redacted]ic sequence:

a State the value of d, the common difference.

b Write a simplified expression for the general term, T_n.

c Find the slope of the line that passes through these points.



17 Each given table of values represents terms in arithmetic sequence. For each table:

i Find d, the common difference.

ii Write a simplified expression for the general term, T_n.

iii Find the missing term in the table.

a

n	1	2	3	4	15
T_n	9	15	21	27	

b

n	1	2	3	4	10
T_n	5	-4	-13	-22	

c

n	1	2	3	4	11
T_n	4	$\frac{17}{2}$	13	$\frac{35}{2}$	

d

n	1	3	5	7	19
T_n	-10	-26	-42	-58	

e

n	1	6	10	18
T_n	-10	-40	-64	

18 The values in the table show terms in an arithmetic sequence for values of n. Complete the table.

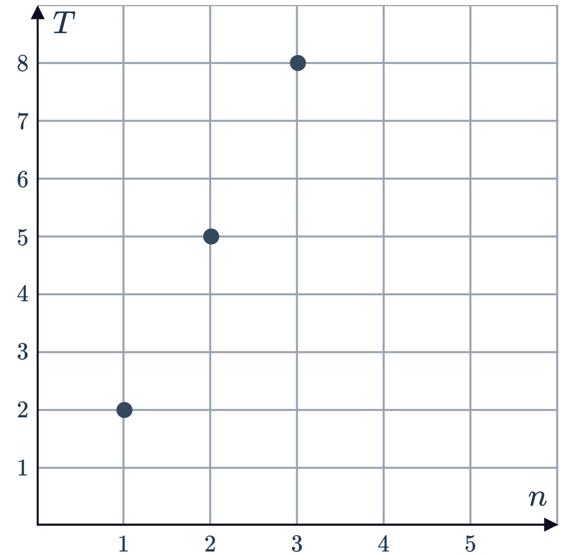
n	1	2	3	4	5
T_n	-6				-26

19 The plotted points represent terms in an arithmetic sequence:

a State the value of d, the common difference.

b Write a simplified expression for the general term, T_n.

c The points are reflected about the horizontal axis to form three new points. If these new points represent consecutive terms of an arithmetic sequence, write the equation for T_k, the kth term in this new sequence.



Applications

20 A scuba diver descends below the surface of the water at a constant rate so that his depth after 4 minutes, 8 minutes and 12 minutes is 15 ft, 30 ft, 45 ft, respectively.
If n is the number of minutes it takes the diver to reach a depth of 120 ft, solve for n.

21 Tobias starts his career with a monthly wage of $\$3500$. At the beginning of each year that follows, he receives a raise and his monthly wage for that year will be $\$160$ greater than the previous year.

a Find his yearly salary in the second year of his service.

b Write down an expression for the total amount earned in m years.

c Solve for the number of years, y, that it would take for him to earn a total of $\$447\,120$.

22 A termite treatment will cost $\$250$ for the first half hour, $\$245$ for the second half hour, $\$240$ for the third half hour and so on. Find the cost of a treatment that takes 6 hours.

23 A paver needs to pave a floor with an area of 800 square meters. He can pave 50 square meters a day.

a Complete the table showing the area left to pave at the start of each day:

b Is this linear or exponential decay?

Day	Area of floor left to pave (m^2)
1	800
2	
3	
4	
5	

24 A telecommunications company sells 1100 cell phones in the first month of its operation. The company plans to increase its sales by 100 cell phones each month.

a How many phones does the company plan to sell in the last month of the 4th year of its operation?

b How many phones does the company plan to sell in the entire 4 year period?

25 A ball starts rolling down a slope. It rolls 25 in during the first second, 53 in during the second second, 81 in during the third second and so on. At this rate, what is the total distance it will have rolled after 10 seconds?

26 When a new school first opened, a students started at the school. Each year, the number of students increases by the same amount, d.
At the beginning of its 7th year, it had 361 students. At the end of the 11th year, the school had 536 students.

a Find d, the number of students who joined the school each year.

b Find a, the number of students that started at the school when it first opened.

c At the end of the nth year, the school has reached its capacity at 921 students. Find the value of n.

27 A worker at a factory is stacking cylindrical-shaped pipes in layers. Each layer contains one pipe less than the layer below it. There are 4 pipes in the topmost layer, 5 pipes in the next layer, and so on. There are n layers in the stack.

a Form an expression for the number of pipes in the bottom layer.

b Find the total number of pipes in the stack in terms of n.

28 For a sprint training exercise, a number of balls are placed in a straight line. The first ball is 4 yd from the start, and there is a 3 yd distance between each of the remaining consecutive balls. There are n balls placed out in the line in total.
Pauline must run from the start, collect the nearest ball and run back to the start, depositing the ball into a box. Then she must run back to collect the next ball and bring it back to the start. She continues this until all n balls have been collected.

- **a** Write an expression for how far Pauline runs to collect and bring back the kth ball.
- **b** Write an expression for how far she runs to collect and deposit all n balls.
- **c** Pauline wants to runs 350 yards in total to collect and deposit all the balls. Solve for n, the number of balls that would need to be placed in a line.

29 Bart is learning to drive. His first lesson is 26 minutes long, and each subsequent lesson is 4 minutes longer than the lesson before.

- **a** How long will his 15th lesson be?
- **b** If Bart reaches 9.6 total hours of driving on his nth lesson, find the value of n.

30 Katrina starts training for a 4.5 km charity trail run by running every week for 28 weeks. She runs 2 km of the course in the first week, and each week after that she runs 250 m more than the previous week. She continues to increase the distance she runs until the week when she runs far enough to complete the course. Each week after that she completes the course without increase.

- **a** How far does she run in the 11th week?
- **b** What is the total distance that Katrina runs in 28 weeks? Round your answer to two decimal places.

31 A fishing trawler spends several days netting crabs. On the first day, it nets 550 lb of crabs, on the next day it nets 538 lb. The amount netted continues to decrease by the same amount each day.

- **a** How many kilograms are netted on the 11th day?
- **b** What is the total weight netted in the first 11 days?
- **c** The trawler returns to port when it has netted a total weight of $10\,400\text{ lb}$. Find the number of days the trawler spent at sea.

32 The balance of a savings account earning simple interest each year is given by the explicit rule

$$V_n = 2200 + 300\,(n-1)$$

where V_n is the balance after n years.

a How much interest is the account earning each year?

b Calculate the account balance after 1 year.

c State the original investment amount.

d Write a recursive rule for V_{n+1} in terms of V_n, and an initial condition V_0.

33 A cell phone depreciates in value by a constant amount per month and its value is given by the explicit rule

$$V_n = 1200 - 20n$$

where V_n is the balance after n months.

a By how much does the value of the phone depreciate each month?

b What was the purchase price of the phone?

c Write a recursive rule for V_{n+1} in terms of V_n, and an initial condition V_0.

34 A diving vessel descends below the surface of the water at a constant rate so that the depth of the vessel after 1 minute, 2 minutes and 3 minutes is 50 m, 100 m and 150 m respectively.

a By how much is the depth increasing each minute?

b What will the depth of the vessel be after 4 minutes?

c Continuing at this rate, what will be the depth of the vessel after 10 minutes?

35 Each day, Kumi withdraws $\$4$ from her bank account to spend on lunch. Before she withdraws this amount on January 1, she has $\$2000$ in her bank account.

a After Kumi withdraws $\$4$ on January 7, calculate the amount left in her bank account.

b How much does Kumi have left in her bank account at the end of February 4?

c Write a recursive rule for T_n in terms of T_{n-1} which defines the amount Kumi has in her account at the end of day n, and an initial condition for T_0.

36 Zuber is a taxi service that charges a $\$1.50$ p[redacted]p fee and $\$1.95$ per mile of travel.

a Find the total charge for a 10 mi journey.

b Write a recursive rule for T_n in terms of T_{n-1} which defines the price of a n mi trip, and an initial condition for T_0.

37 The value of Beirut Bank shares is decreasing by $\$2.05$ each day. At the beginning of today's trading, the shares are worth $\$42.54$.

a Today is March 8. How much are they worth at the start of March 17?

b Write a recursive rule for V_{n+1} in terms of V_n which defines the the value of the shares at the end of day n, and an initial condition for V_0.

38 Tabitha is a salesperson and is paid a travel allowance of $\$45$ for each business trip she takes to demonstrate the use of the product she sells. Each week she is also paid a retainer of $\$230$.

a Calculate the total amount she is paid in a week where she does three business trips.

b Write a recursive rule for u_n in terms of u_{n-1} which defines how much Tabitha is paid in a week where she makes n business trips, and an initial condition for u_0.

c Write a recursive rule for v_n in terms of v_{n-1} which defines how much Tabitha is paid in two week where she makes n business trips, and an initial condition for v_0.

39 For a fibre-optic cable service, Christa pays a one off amount of $\$200$ for installation costs and then a monthly fee of $\$30$.

a Complete the table of values for the total cost (T) of Christa's service over n months.

n	1	2	3	4	18
T					

b By how much are consecutive terms in the sequence increasing?

c Considering the table of values, plot the points corresponding to $n = 1, 2, 3$ and 4.

d If the points on the graph were joined, would they form a straight or curved line?